

RENANCE

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# FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MEE 212 — Past Questions & Materials  
**by Renance**

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Course: MEE 212  
Institution: Federal University of Petroleum Resources, Effurun  
Papers in this pack: 4  
Levels: 200

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This study pack was compiled and restructured by Renance for MEE 212 students at Federal University of Petroleum Resources, Effurun. It brings together 4 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

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## Paper 1 of 4 — MEE 212 2023 PQ

Level 200 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN  
DEPARTMENT OF MECHANICAL ENGINEERING  
First Semester Examination 2022/2023 Session  
COURSE CODE / TITLE: MEE 212 ENGINEERING MECHANICS I

TIME ALLOTTED: 2 HOURS

INSTRUCTION: ANSWER QUESTION ONE AND ANY OTHER TWO QUESTIONS

**QUESTION 1**

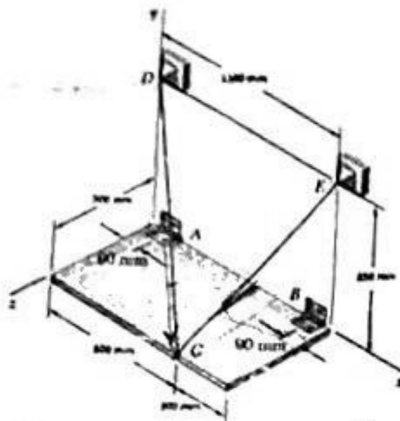


Figure Q1 (30 marks)

The uniform rectangular plate is supported in the position shown by hinges A and B and by cable DCE that passes over a frictionless pulley at C. The tension in the string on path DC is 1200 N, while that on the part of EC is 1500 N. Neglecting the weight of the plate and reactions at the hinges, determine the following:

- The rectangular components of both tensions
- The magnitude of singular force that can be used to replace the entire string (Resultant of DC and EC)
- The moment of both tensile forces about A.
- The resultant moment about A.

**QUESTION 2**

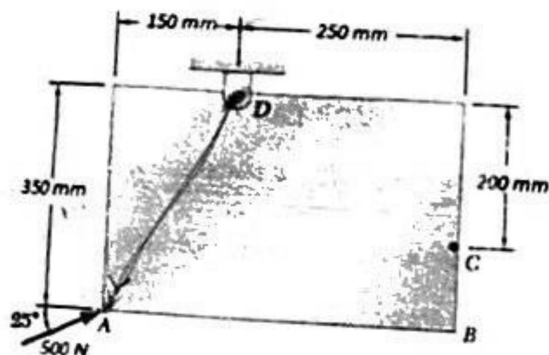
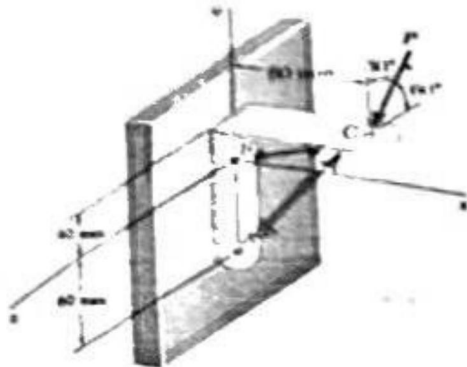


Figure Q2 (20 marks)

For Figure Q2, determine

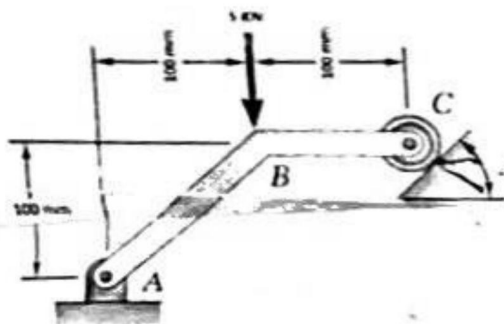
- The moment of the 500 N force about D
- The magnitude and direction of a horizontal force that would be applied at point C to create the same moment about D.

**QUESTION 3**

For Figure Q3, the moment of the force  $P$  about point  $A$  is  $3.5 \text{ kN.m}$ , determine:

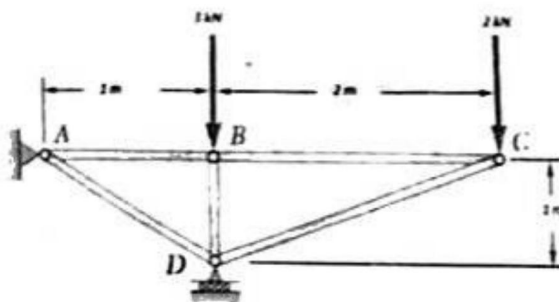
- The magnitude of the force  $P$ .
- The moment of the force about point  $B$ .

Figure Q3 (20 marks)

**QUESTION 4**

A force of  $5 \text{ kN}$  is applied to a bent rod  $ABC$ , which is supported as shown in Figure Q4. Determine the reactions at the support

Figure Q4 (20 marks)

**QUESTION 5**

The truss, used to support a balcony, is subjected to the loading shown in Figure Q5. Approximate each joint as a pin and determine the force in each member, and state whether the members are in tension or compression.

Figure Q5 (20 marks)

## Paper 2 of 4 — MEE 212 PQ

Level 200 • First Semester • PastQuestion

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MEE212 practice problems

QUESTION 1

QUESTION 2

QUESTION 3

QUESTION 4

QUESTION 5

QUESTION 6

QUESTION 7

QUESTION 8

Find the moment of the 150N force about A

## Paper 3 of 4 — MEE 212 Past Questions (2018/2019)

Level 200 • First Semester • PastQuestion

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## FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

## DEPARTMENT OF MECHANICAL ENGINEERING

## EXAMINATIONS

1<sup>st</sup> Semester 2018/2019

B.Eng. Mechanical Engineering

April 2019

MEE 212 Engineering Mechanics I

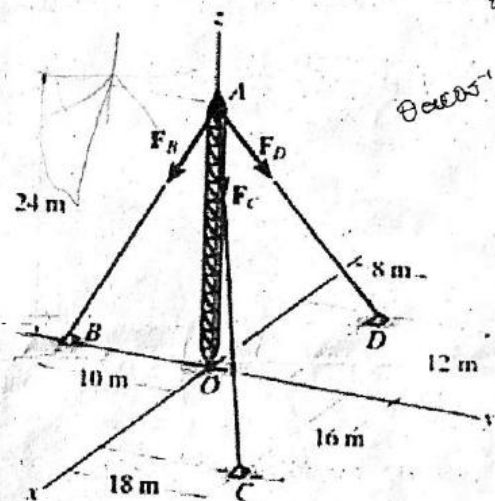
Answer any THREE of the FIVE questions.

Time allowed: 2Hrs 15mins

**Question 1**

The antenna tower is supported by three cables. If the forces of these cables acting on the antenna are  $F_B = 520\text{ N}$ ,  $F_C = 680\text{ N}$ , and  $F_D = 560\text{ N}$ , determine the magnitude and the coordinate direction angles of the resultant force acting at A.

Fig. Q.1

**Question 2**

A vertical force of 890 N is required to remove the nail at C from the board by means of a bent crowbar ABC, Fig. Q.2. As the nail first starts moving, determine

- the moment about B of the force exerted on the nail,
- the magnitude of the force P which creates the same moment at B if  $\alpha = 10^\circ$ ,
- the smallest force P which creates the same moment about B

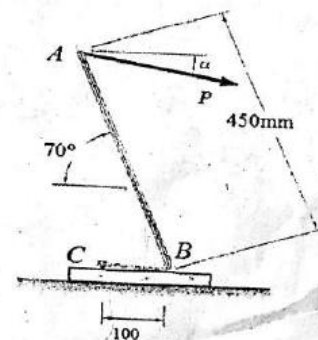


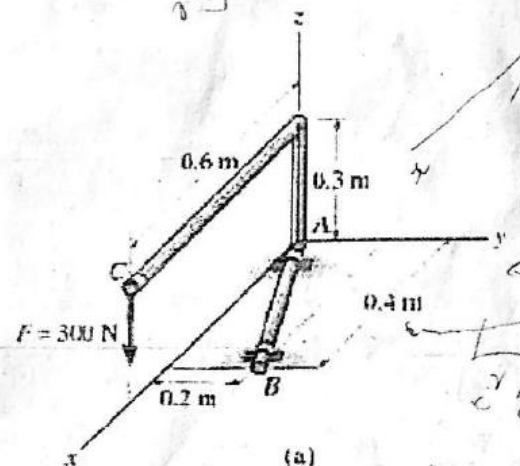
Fig. Q.2

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**Question 3**

- Determine the moment  $M_{AB}$  produced by the force  $F$  in Fig. Q.3 which tends to rotate the rod about the  $AB$  axis.
- State with reasons whether it is possible or not for the force to rotate the rod about  $z$  axis, Fig Q.3.

Fig. Q.3

**Question 4**

- State the conditions of equilibrium for a rigid body in space.
- The 14-N circular drainage pipe (shown in cross-section) is held in the tines of the fork lift. Determine the normal forces at A and B as functions of the blade angle.

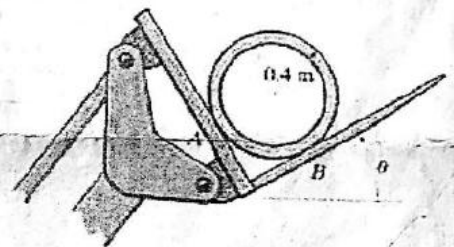
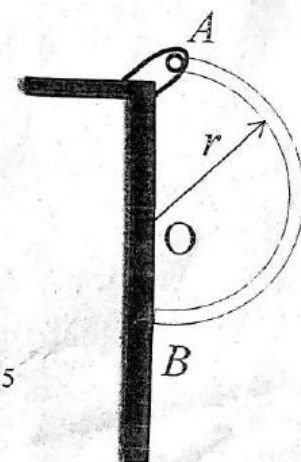


Fig. Q.4

**Question 5**

A uniform semicircular rod of weight  $W$  and radius  $r$  is attached to a pin at A and rests against a frictionless surface at B, Fig. Q.5. Determine the reactions at A and B.

Fig. Q.5



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## Paper 4 of 4 — MEE 212 PAST QUESTION AND PRACTICE QUESTION

Level 200 • First Semester • PastQuestion

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MEE212 practice problems

QUESTION 1

QUESTION 2

QUESTION 3

QUESTION 4

QUESTION 5

QUESTION 6

QUESTION 7

QUESTION 8

Find the moment of the 150N force about A

MEE212 practice problems

QUESTION 1

QUESTION 2

QUESTION 3

QUESTION 4

QUESTION 5

QUESTION 6

QUESTION 7

QUESTION 8

Find the moment of the 150N force about A

MEE212 practice problems

QUESTION 1

QUESTION 2

QUESTION 3

QUESTION 4

QUESTION 5

QUESTION 6

QUESTION 7

QUESTION 8

Find the moment of the 150N force about A